

Social choice

1. A Condorcet winner is an alternative that ...
 - (a) has the most first choice votes
 - (b) is ranked last by no one
 - (c) has the highest Borda score
 - (d) beats every other alternative in a pairwise majority vote
2. A Condorcet cycle is when ...
 - (a) the Borda winner is always unique
 - (b) pairwise majority preferences cycle even when every voter's ranking is transitive
 - (c) simple majority rule always picks the Condorcet winner
 - (d) voters cannot rank candidates
3. Borda's method scores each alternative by ...
 - (a) the number of alternatives it is ranked above, summed over all voters
 - (b) whether it beats every rival head-to-head
 - (c) a coin toss
 - (d) the number of first choice votes only
4. Arrow's impossibility theorem states that ...
 - (a) simple majority rule is always transitive
 - (b) Borda's method is the only fair voting rule
 - (c) no social welfare function satisfies unrestricted domain, Pareto efficiency, independence of irrelevant alternatives and non-dictatorship together
 - (d) every preference profile has a Condorcet winner
5. Simple majority rule ranks x above y when ...
 - (a) a strict majority of voters prefer x to y
 - (b) x has a higher Borda score than y
 - (c) any voter prefers x to y
 - (d) x has more first choice votes than y
6. A social welfare function maps ...
 - (a) every preference profile to a collective ranking of the alternatives
 - (b) candidates to vote totals only
 - (c) payoffs to strategies
 - (d) each voter to a single candidate

7. A preference profile is ...
- (a) the list of every voter's ranking of the alternatives
 - (b) the winning candidate
 - (c) a single voter's top choice
 - (d) a payoff matrix
8. Each voter's preference is assumed to be ...
- (a) a transitive strict ranking of the alternatives
 - (b) a probability distribution
 - (c) a single favourite only
 - (d) possibly cyclic
9. Independence of irrelevant alternatives requires that the collective ranking of x and y ...
- (a) always matches the Borda ranking
 - (b) is decided by a single voter
 - (c) ignores how voters rank x and y
 - (d) depends only on how voters rank x against y , not on any third alternative
10. When a Condorcet winner exists, it is ...
- (a) one of several
 - (b) unique
 - (c) never the Borda winner
 - (d) always the Borda winner
11. Borda's method always ...
- (a) produces a complete ranking and never cycles
 - (b) fails when preferences differ
 - (c) selects the Condorcet winner
 - (d) ignores lower preferences
12. The Borda winner ...
- (a) never exists
 - (b) need not be the Condorcet winner, even when one exists
 - (c) is always the simple majority winner
 - (d) is always the Condorcet winner
13. A Condorcet cycle has the form ...

- (a) $A \succ B \succ C$ with no return
 - (b) everyone ranking A first
 - (c) $A \succ B \succ C \succ A$ under pairwise majority
 - (d) a tie between all candidates
14. Condorcet's method can fail to select a winner because ...
- (a) voters cannot rank candidates
 - (b) it always produces a tie
 - (c) it ignores first choice votes
 - (d) pairwise majorities may cycle, so no candidate beats all others
15. Different voting methods applied to the same profile ...
- (a) can select different winners
 - (b) are forbidden
 - (c) always tie
 - (d) always agree
16. A voter's preference function is a strict linear order, meaning it is asymmetric, transitive and ...
- (a) reflexive
 - (b) random
 - (c) cyclic
 - (d) complete (every pair of distinct alternatives is ranked)
17. Arrow's impossibility theorem applies to sets of alternatives of size ...
- (a) exactly 2
 - (b) at least 10
 - (c) at least 3
 - (d) any size, including 1
18. Both Borda's method and Condorcet's method respond to Arrow's theorem by giving up which property?
- (a) independence of irrelevant alternatives
 - (b) unrestricted domain
 - (c) non-dictatorship
 - (d) Pareto efficiency
19. With three alternatives, Borda's method awards a voter's top choice ...

(a) 2 points

(b) 3 points

(c) 1 point

(d) 0 points

20. A voting rule is open to strategic manipulation when a voter can gain by ...

(a) adding a new candidate

(b) abstaining from the vote

(c) ranking alternatives honestly

(d) misreporting their true preferences